Huawei Access Node JAY-ACC-13 QoS Congestion Event ALARM_5765

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5765

Description

The QoS Congestion Event failure domain on the Huawei Access Node JAY-ACC-13 at the Jayanagar Metro Site occurs due to microbursts, interface tail drops, and QoS queue starvation. This is a complex phenomenon resulting from the interplay of various factors, including but not limited to, packet classification, queuing mechanisms, and traffic policing. In a Huawei Access Node, the QoS framework is implemented using the Hierarchical Queuing (HQ) mechanism, which classifies traffic into different queues based on packet attributes such as DSCP, IP precedence, and 802.1p priority. When the HQ mechanism is overwhelmed by a burst of traffic, it may lead to queue starvation, causing microbursts and interface tail drops. This, in turn, triggers the QoS Congestion Event alarm (ALARM_5765) on the Huawei Access Node.

Pre-requisite Checks

- Pre-requisite: Verify that the Jayanagar Metro Site is accessible and the Huawei Access Node JAY-ACC-13 is powered on.

- Pre-requisite: Check the redundant link status to ensure that there are no single points of failure.

- Pre-requisite: Gather relevant logs and performance data to aid in troubleshooting.

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops due to HQ queue starvation	High
Incorrect QoS configuration	Medium
Traffic policing issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface GigabitEthernet 0/0/1, Expected Output: Interface GigabitEthernet 0/0/1 is up (line protocol is up), Description: This command checks the status of the GigabitEthernet 0/0/1 interface, verifying that it is up and operational.

Command: display qos queue GigabitEthernet 0/0/1, Expected Output: Queue ID : 1, Queue Name : HQ1, Queue Threshold : 80%, Description: This command displays the QoS queue configuration for the GigabitEthernet 0/0/1 interface, including the queue ID, name, and threshold.

Command: display traffic policing GigabitEthernet 0/0/1, Expected Output: Traffic Policing : enabled, Policing Rate : 1000000,
```

```
Description: This command checks the traffic policing configuration for the GigabitEthernet 0/0/1 interface, verifying that it is enabled and set to the correct policing rate.
```

Resolution Steps

Step 1: Step: Verify QoS configuration, Command: display qos config GigabitEthernet 0/0/1, Description: Verify that the QoS configuration is correct and not causing the congestion event.

Step 2: Step: Adjust HQ queue threshold, Command: queue threshold GigabitEthernet 0/0/1 queue-id 1 70%, Description: Adjust the HQ queue threshold to a lower value to prevent queue starvation.

Step 3: Step: Adjust traffic policing rate, Command: traffic policing GigabitEthernet 0/0/1 policing-rate 900000, Description: Adjust the traffic policing rate to a lower value to prevent over-policed traffic.

Post-Check Verification

```
Command: display interface GigabitEthernet 0/0/1, Expected Output: Interface GigabitEthernet 0/0/1 is up (line protocol is up), Description: Verify that the interface is up and operational after adjusting the QoS configuration.

Command: display qos queue GigabitEthernet 0/0/1, Expected Output: Queue ID : 1, Queue Name : HQ1, Queue Threshold : 70%, Description: Verify that the QoS queue configuration has been adjusted correctly.

Command: display traffic policing GigabitEthernet 0/0/1, Expected Output: Traffic Policing : enabled, Policing Rate : 900000, Description: Verify that the traffic policing configuration has been adjusted correctly.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: queue threshold GigabitEthernet 0/0/1 queue-id 1 80%, Description: Restore the original QoS configuration to prevent any further issues.

- Step: Restore original traffic policing rate, Command: traffic policing GigabitEthernet 0/0/1 policing-rate 1000000, Description: Restore the original traffic policing rate to prevent any further issues.